

# Samsung Innovation Campus

## **Chapter 9.** Deep Learning Module

# Vision and Language

(Plus the CLIP Model)

(and SigLiP as a bonus)

# In the latest episode of the Deep Learning Module...

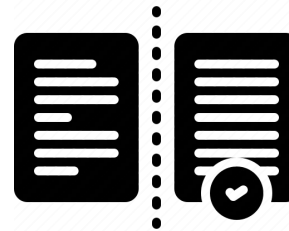
You saw the encoder architecture and how it can be used for many tasks in Natural Language Processing (NLP).



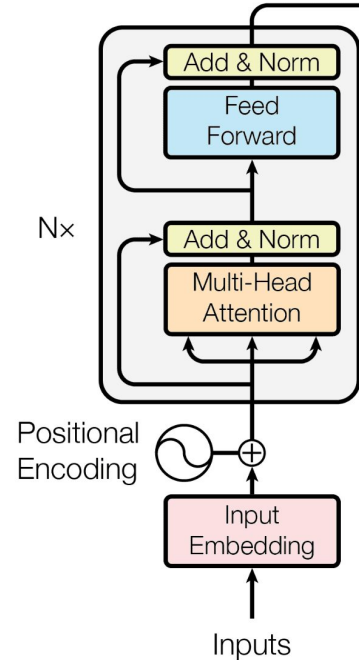
Retrieval



Sentiment



Similarity



**You even got to meet BERT!**



# Episode 2: What about Vision?

Vision tasks have **different requirements**:

1. **Complex** visual information: We need to understand individual objects, **scenes**, and **compositions**

2. **Global Context**: We need to capture **local** and **long-range dependencies**

3. **Generalization** across different image variations and different tasks  
“A black cat on a hardwood floor inside a house.”



# Episode 3: What about Vision and Languages?

Vision and Language (V&L) tasks are crucial for models to understand the world in a more “human-like” way.

**If humans naturally combine both modalities, why shouldn't models do the same?**

**Visual  
Question-Answering**



**Image Captioning**



**Cross-modal Retrieval**



# V&L Tasks: Visual Question-Answering

**Input:** Image and a question



What is this museum?

**Output:** Answer

This can either be similar to a **multiple question** test - where the model knows all the possible answers *a priori* - or **generated!**

# V&L Tasks: Image Captioning

**Input:** Image



Besides simply describing the image, this can be used for **search**: where the query *is* the image description and the retrieved documents may be either images or text

**Output:** Caption

A picture of the MAAT museum in Lisbon, taken at night

# V&L Tasks: Cross-modal Retrieval

**Input:** Image



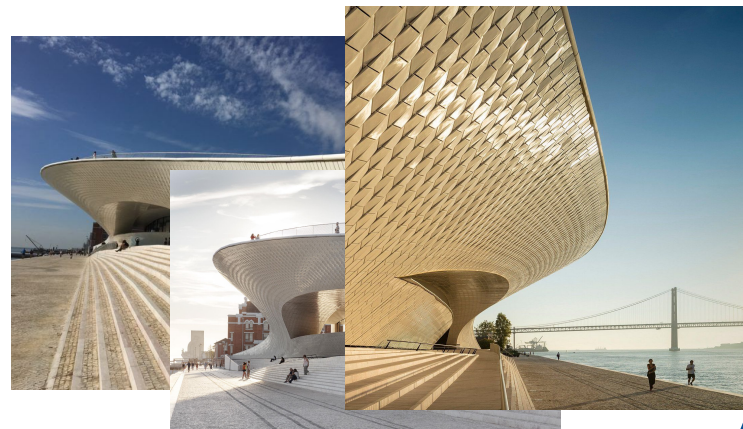
**Output:** Text



**Input:** Text

maat museum

**Output:** Image





# V&L Models: How?

Visual and Language Models are a collection of models that can learn both from **textual** and **visual** information.

Although these models are often **generative** - capable of generating captions or an image, we first need to focus on how can these models...

... understand **text**?

Text Encoder

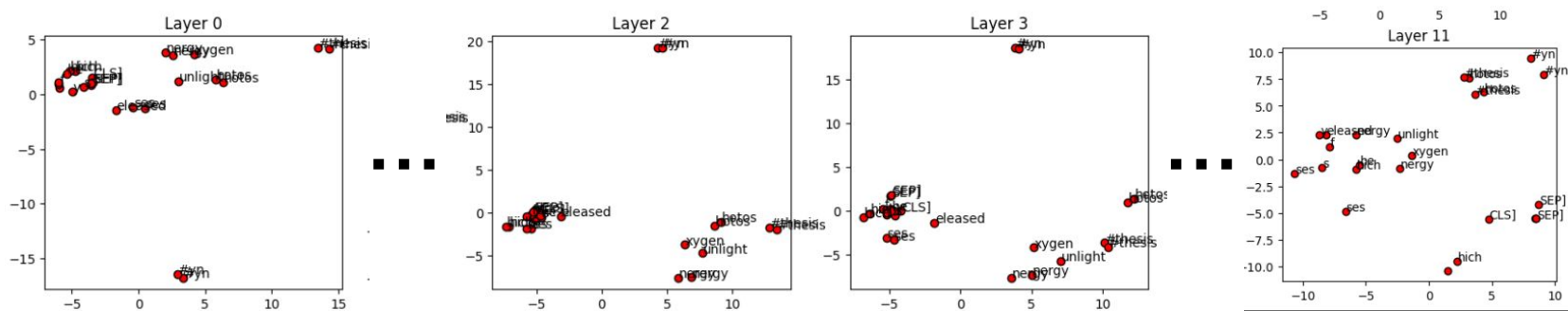
... understand **images**?

Vision Encoder

# Model Input: Text

Text is encoded as word embeddings plus their positional embeddings.

Its is then processed with the usual pipeline in order to obtain the **contextual embeddings**.



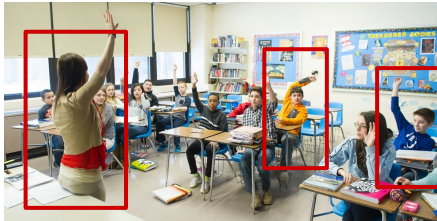
# Model Input: Image

Are there visual embeddings?

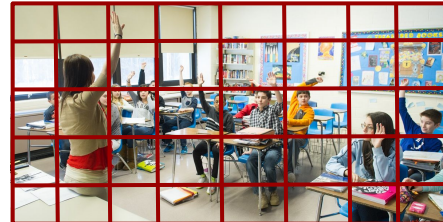
**Yes!** We need vector representations of images in order to represent them on the **same** higher dimensional space as text.

How can we get them?

## Region-based Visual Embeddings



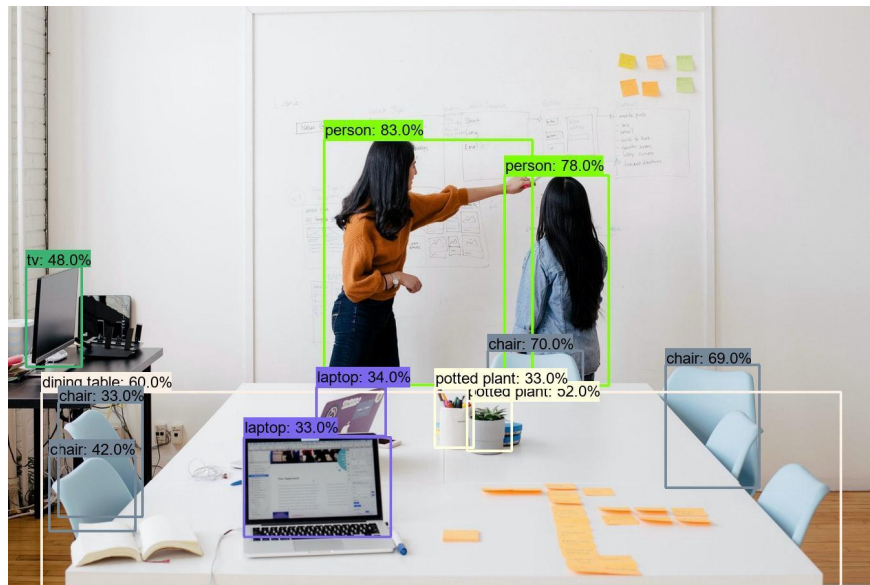
## Grid-based Visual Embeddings



# Region-based Embeddings

Image region features:

- Bounding boxes
- Visual features extracted from a pre-trained object detection model



This can be achieved with CNNs!

# Grid-based Embeddings

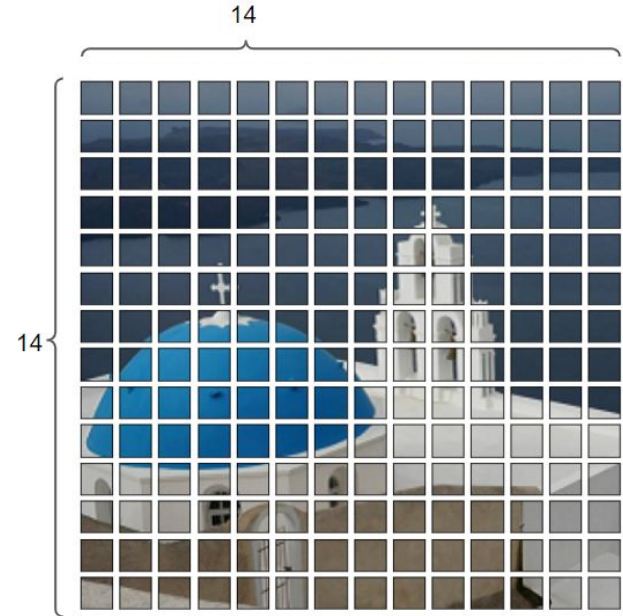
**Goal:** Divide an image into a uniform grid and extract individual embeddings for each square (**patch**).

1. The image is **center cropped** to a **224x224** pixel image

Enough detail is retained

Training is efficient

2. Create a **14x14** grid



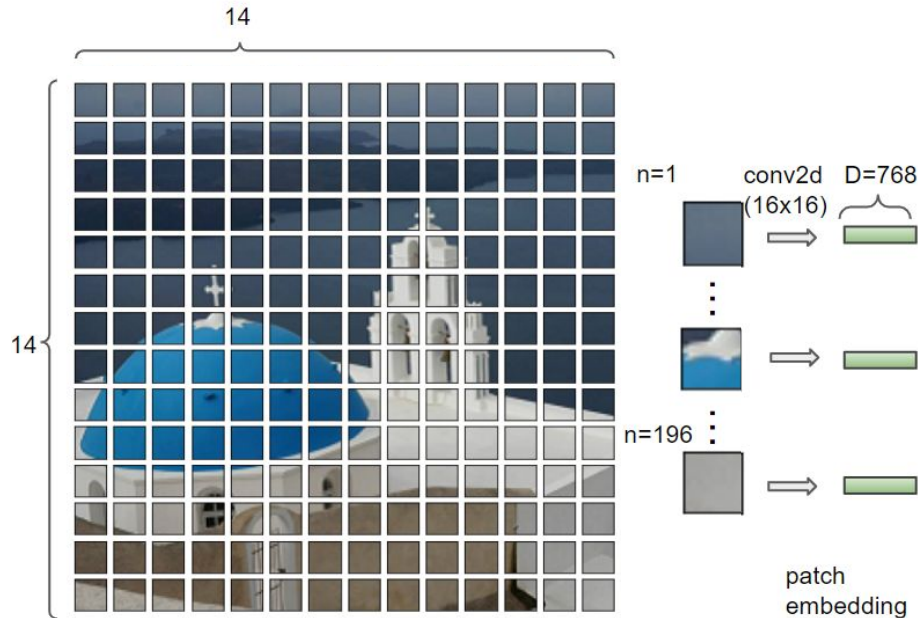
# Grid-based Embeddings

Assuming a  $14 \times 14$  grid, each patch will be  **$16 \times 16$**  pixels.

**Total n° of patches:** 196

**Total of pixels per patch:** 768  
(256 per channel in an RGB image)

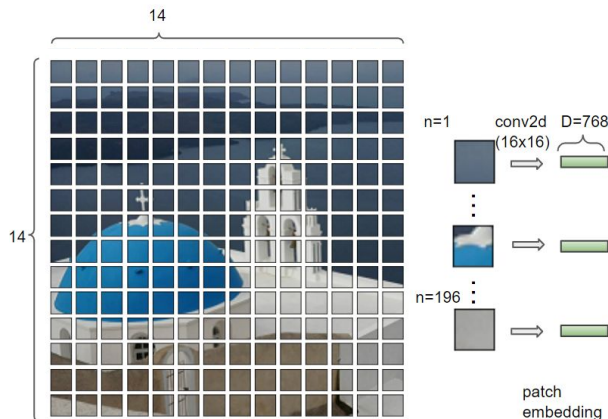
**3. A transformation** is applied to each patch to obtain a single 768 dimensional vector



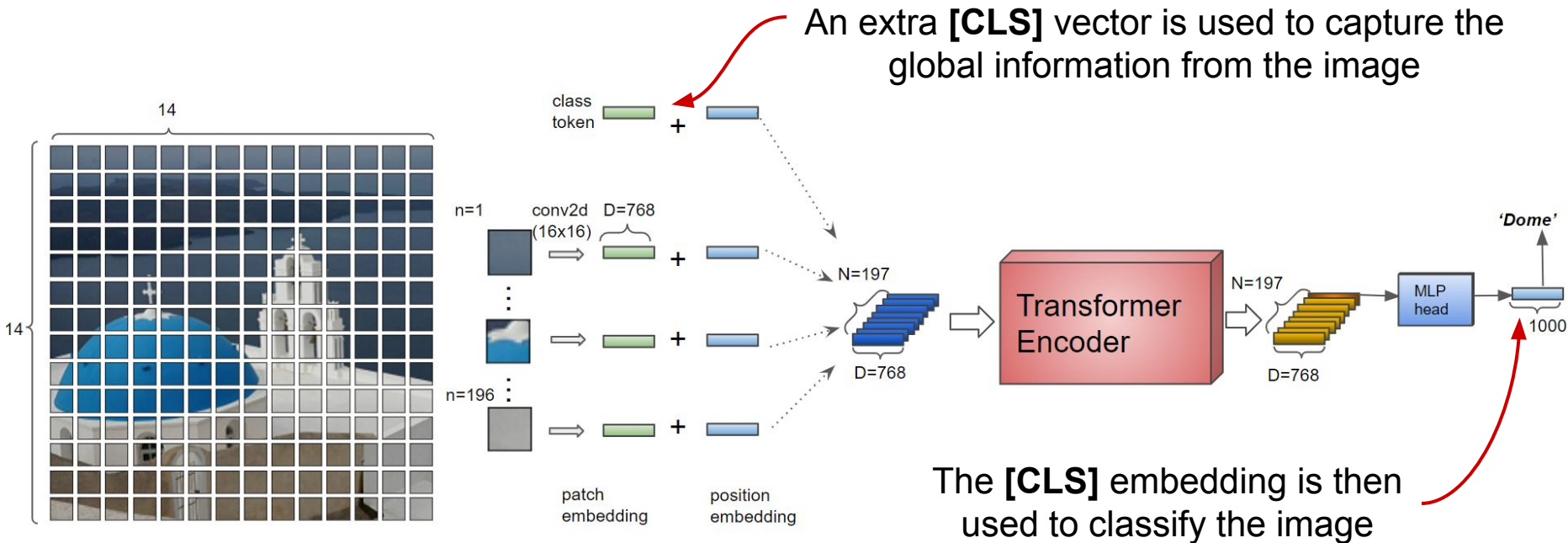
# Grid-based Embeddings

This transformation can be:

- **Convolutional Transformation:** The patches are processed to obtain information about the edges, textures, and patterns
- **Linear Transformation:** Flattens the image embeddings into a vector

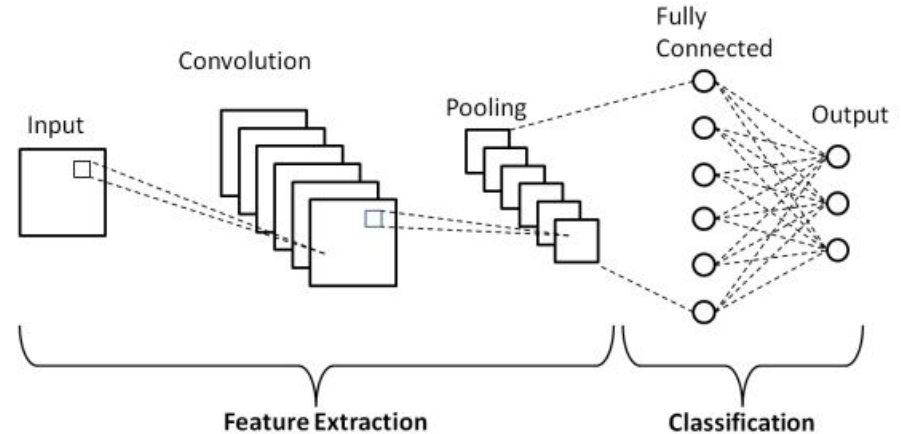
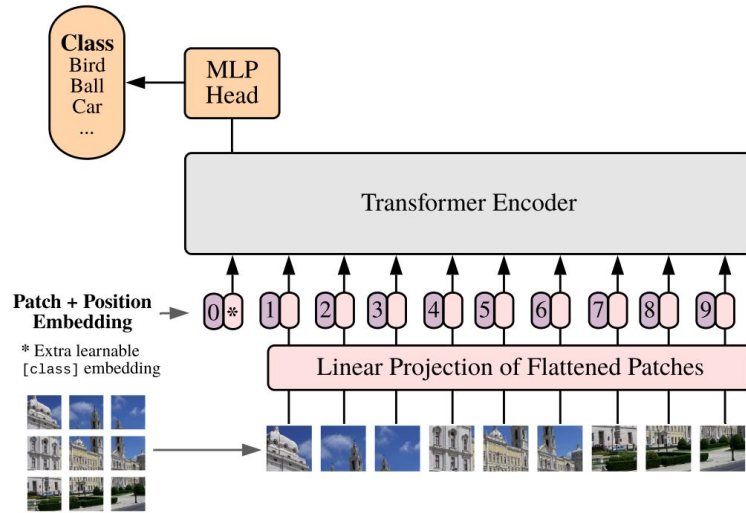


# Vision Transformer (ViT)





# Did ViT overthrow CNNs? Not quite..



**Both of these architectures co-exist!**

Both with their respective advantages and disadvantages

# ViT vs. CNNs: The Showdown

The choice depends on **dataset size**, **computational resources**, and **task complexity**.

## ViT

### Advantages

Captures **global-context**

**Few** inductive biases

### Disadvantages

Computationally **expensive**

Data hungry

## CNN

### Advantages

Computationally **efficient**

Smaller datasets

### Disadvantages

Global context is **limited**

Limited flexibility

# ViT & Multimodal Tasks

It is very simple to leverage ViT for multimodal tasks since both vision and text quickly **align within the same embedding space** due to similarities in architecture and data representation.

How do we combine both modalities?



Fusion Techniques

Single Modality  
(no fusion)

Single-stream Transformer  
(early fusion)

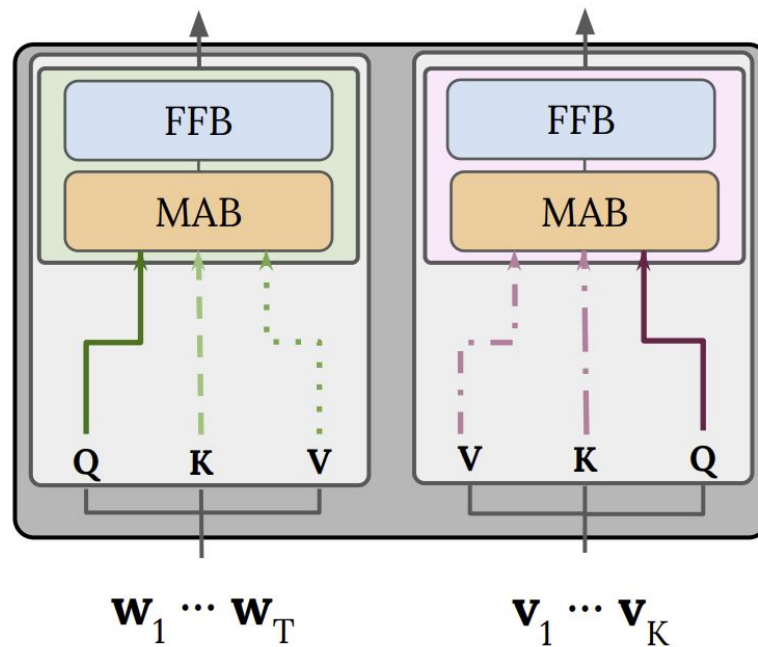
Dual-stream Transformer  
(late fusion)

# Dual Encoders

One **encoder** per **modality**!

Each modality is processed **independently** leading to **text** and **vision** embeddings being computed **separately**

The embeddings are only aligned in a **shared embedding space**!

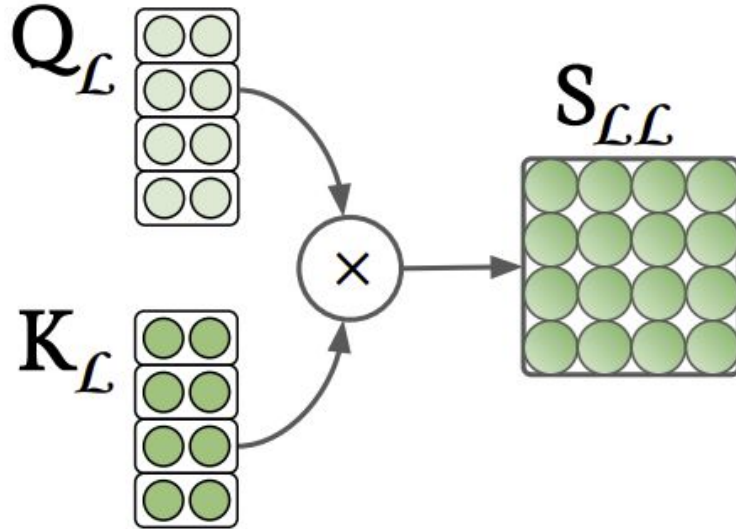


MAB = Multi-Head Attention Block

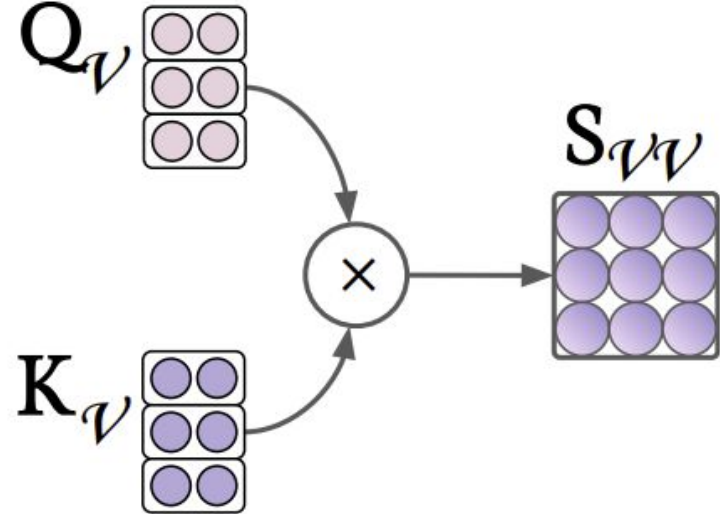
FFB = Feed-Forward Block

# Dual Encoders: Self-Attention

## Text Self-Attention

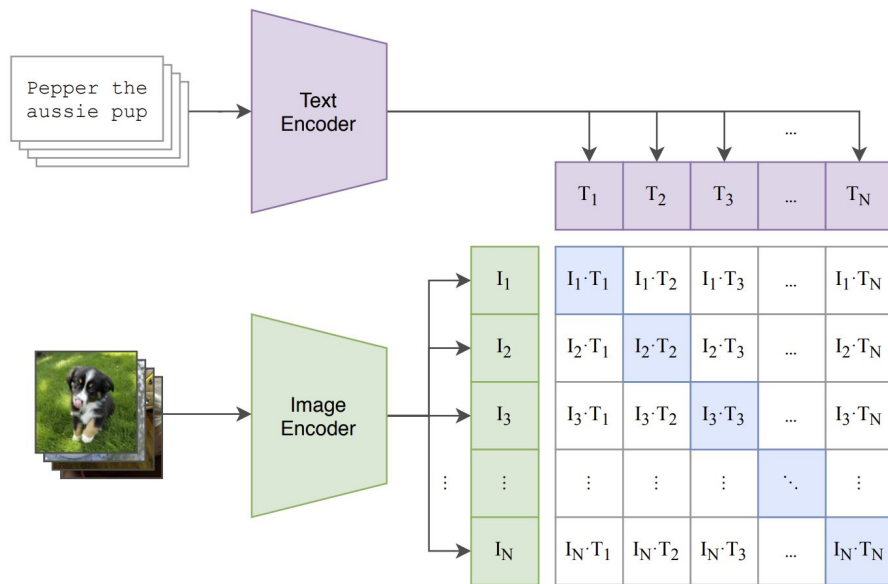


## Vision Self-Attention



# CLIP: Contrastive Language-Image Pre-training

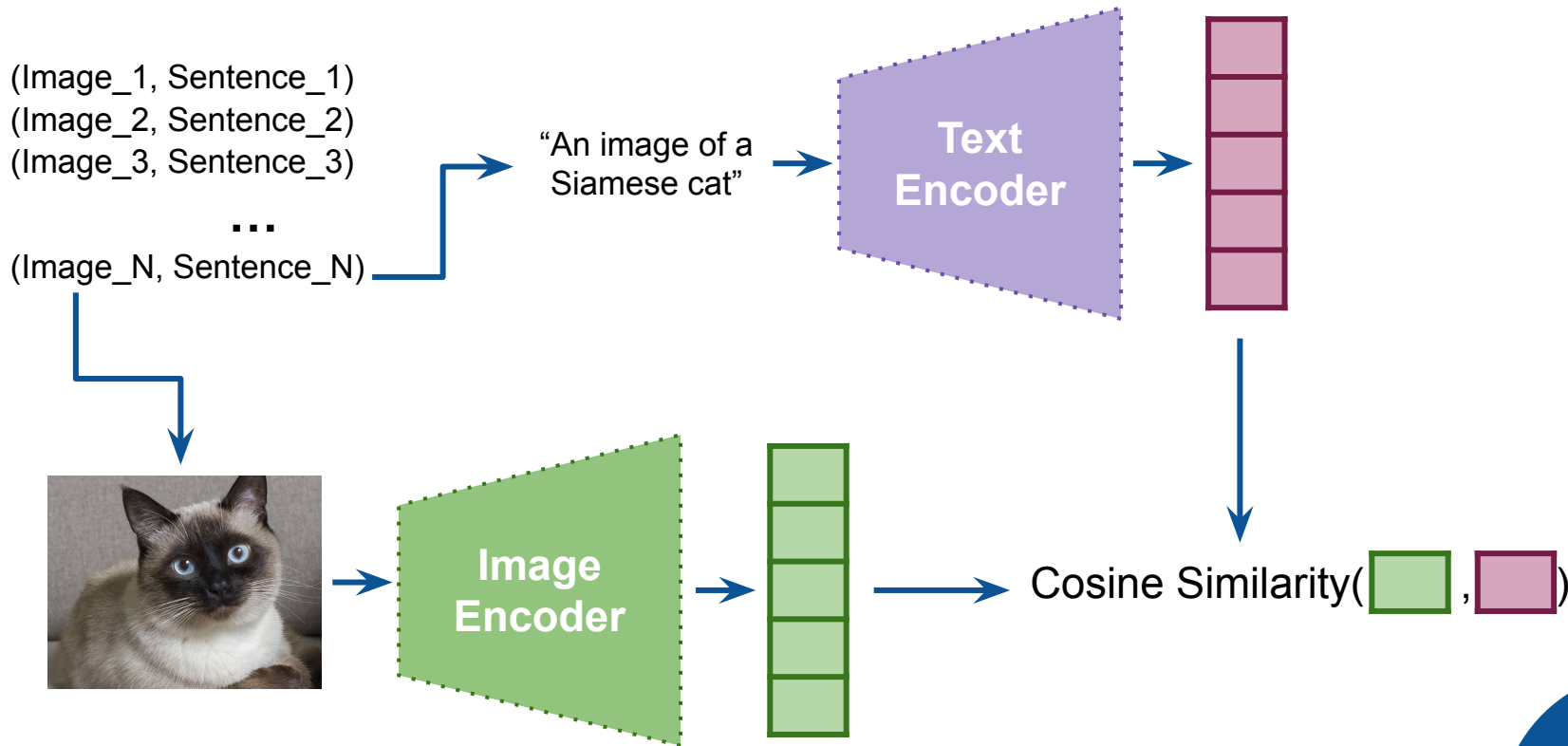
Multimodal model that aligns visual and textual embeddings in a shared semantic space using a **contrastive learning** objective.



Trained on **400 million** image/sentence pairs: 1 image-sentence pair/second for 12 days!

1. **Two encoders**, one per modality
2. **Independent** embedding computation
3. **Shared** embedding space

# CLIP: Two independent encoders



# CLIP: Contrastive Learning Objective

This model learns by **maximizing** the similarity of **correct** image-text pairs while **minimizing** the similarity of **incorrect** pairs.

|          | $T_1$           | $T_2$           | $T_3$           | ...      | $T_N$           |
|----------|-----------------|-----------------|-----------------|----------|-----------------|
| $I_1$    | $I_1 \cdot T_1$ | $I_1 \cdot T_2$ | $I_1 \cdot T_3$ | ...      | $I_1 \cdot T_N$ |
| $I_2$    | $I_2 \cdot T_1$ | $I_2 \cdot T_2$ | $I_2 \cdot T_3$ | ...      | $I_2 \cdot T_N$ |
| $I_3$    | $I_3 \cdot T_1$ | $I_3 \cdot T_2$ | $I_3 \cdot T_3$ | ...      | $I_3 \cdot T_N$ |
| $\vdots$ | $\vdots$        | $\vdots$        | $\vdots$        | $\ddots$ | $\vdots$        |
| $I_N$    | $I_N \cdot T_1$ | $I_N \cdot T_2$ | $I_N \cdot T_3$ | ...      | $I_N \cdot T_N$ |

We use the similarity scores to obtain two probability distributions: one for the **text-to-image (T2I)** and one for **image-to-text (I2T)**

Softmax

1. Compute Cross-entropy loss **T2I**
2. Compute Cross-entropy loss **I2T**
3. **Average** them!



# CLIP: Zero-Shot Capabilities

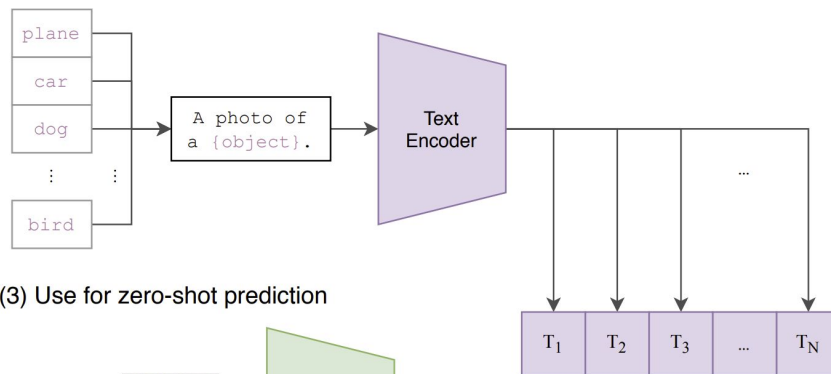
This model **generalizes** extremely well in tasks for which it was never trained for (**zero-shot prediction**).

**Image annotation** can be achieved with prompts like:

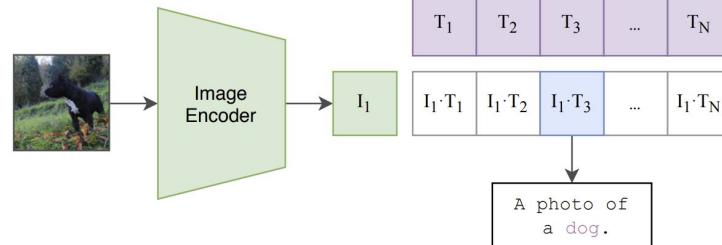
A photo of a \_\_\_\_\_.

It can also be used for **image-to-text** and **text-to-image** retrieval.

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction



# CLIP: Pre-training and Data

A major motivation for natural language supervision is the large quantities of data of this form available publicly on the internet. Since existing datasets do not adequately reflect this possibility, considering results only on them would underestimate the potential of this line of research. To address this, we constructed a new dataset of 400 million (image, text) pairs collected from a variety of publicly available sources on the Internet. To attempt to cover as broad a set of visual concepts as possible, we search for (image, text) pairs as part of the construction process whose text includes one of a set of 500,000 queries.<sup>1</sup> We approximately class

their local batch of embeddings. The largest ResNet model, RN50x64, took 18 days to train on 592 V100 GPUs while the largest Vision Transformer took 12 days on 256 V100 GPUs. For the ViT-L/14 we also pre-train at a higher 336 pixel resolution for one additional epoch to boost performance similar to FixRes (Touvron et al., 2019). We denote this model as ViT-L/14@336px. Unless otherwise specified, all results reported in this paper as “CLIP” use this model which we found to perform best.

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<sup>1</sup>The base query list is all words occurring at least 100 times in the English version of Wikipedia. This is augmented with bi-grams

# CLIP: Fine-tuning Magic

## BioCLIP: A Vision Foundation

## ANOMALYCLIP: OBJE ING FOR ZERO-SHOT A

Qihang Zhou<sup>1\*</sup>, Guansong Pang<sup>2\*</sup>, Yu  
<sup>1</sup>Zhejiang University <sup>2</sup>Singapore Man  
<sup>1</sup>{zqhang, s18he, cjm}@zju.e  
<sup>3</sup>ytian11@meei.harvard.edu

## MoralCLIP: Contrastive Alignment of Vision-and-Language Representations with Moral Foundations Theory

Ana Carolina Condez  
NOVA LINGS, NOVA School of  
Science and Technology  
Caparica, Portugal  
a.condez@campus.fct.unl.pt

Diogo Tavares  
NOVA LINGS, NOVA School of  
Science and Technology  
Caparica, Portugal  
dc.tavares@campus.fct.unl.pt

João Magalhães  
NOVA LINGS, NOVA School of  
Science and Technology  
Caparica, Portugal  
jmag@fct.unl.pt

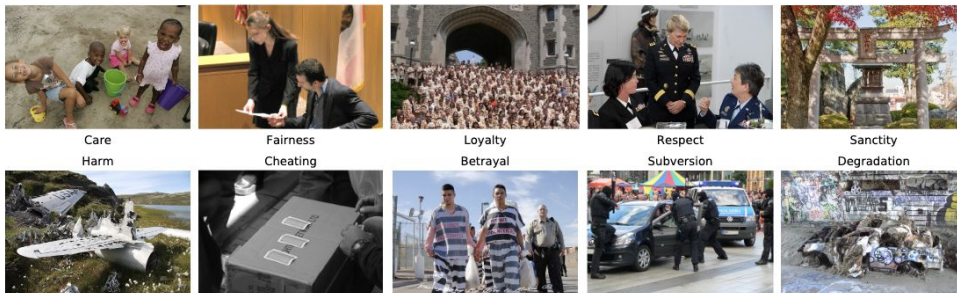


Figure 1: Visual depictions of fundamental moral dimensions as defined by Moral Foundations Theory (MFT) [13]. Images were extracted from the SMID [7] dataset, which was annotated at large-scale by humans and validated by experts. MoralCLIP successfully creates a vision and language embedding space where these dimensions are well captured.

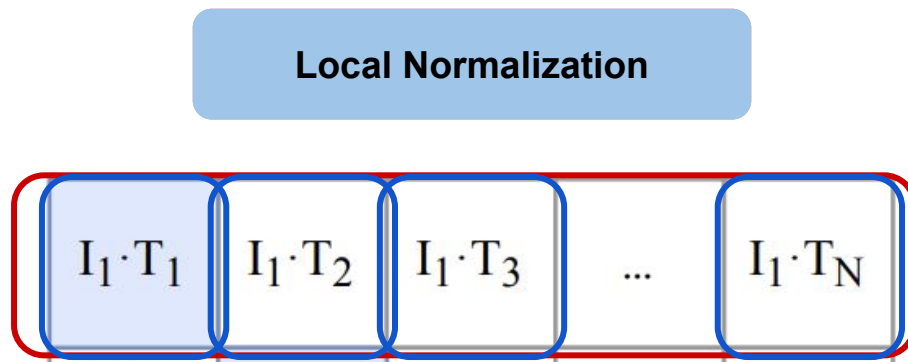
Abstract

CCS Concepts

# SigLIP: The New Age

**SigLIP** (Sigmoid Loss for Language-Image Pre-training) replaces the softmax loss with a **pairwise sigmoid loss**.

**Operates independently for each image-pair!**

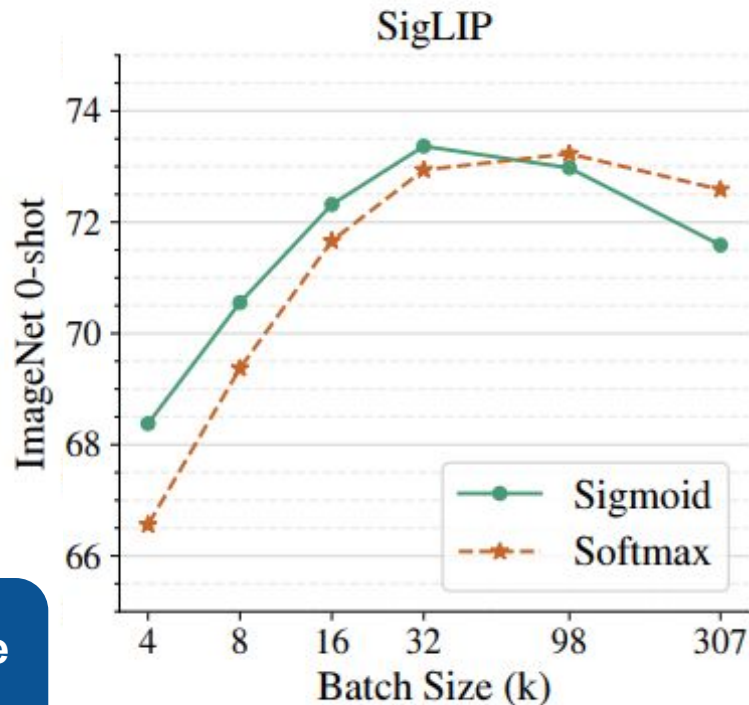


# SigLIP: Why?

This is a more efficient alternative, reducing the need for such intense computational resources, as the loss is now computed over the **entire similarity matrix**

Performs **comparably or better** than softmax, especially at smaller batch sizes

**Scales more efficiently as batch size increases**



# SigLIP vs CLIP: Loss Functions

## CLIP: SoftMax

$$-\frac{1}{2|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \left( \overbrace{\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_i \cdot \mathbf{y}_j}}}^{\text{image} \rightarrow \text{text softmax}} + \overbrace{\log \frac{e^{t\mathbf{x}_i \cdot \mathbf{y}_i}}{\sum_{j=1}^{|\mathcal{B}|} e^{t\mathbf{x}_j \cdot \mathbf{y}_i}}}^{\text{text} \rightarrow \text{image softmax}} \right)$$

## SigLIP: Sigmoid

$$-\frac{1}{|\mathcal{B}|} \sum_{i=1}^{|\mathcal{B}|} \sum_{j=1}^{|\mathcal{B}|} \underbrace{\log \frac{1}{1 + e^{z_{ij}(-t\mathbf{x}_i \cdot \mathbf{y}_j + b)}}}_{\mathcal{L}_{ij}}$$

# SigLIP: The proof is in the pudding

## SigLIP (2023)

|                     | Image | Text | BS   | #TPUv4 | Days |
|---------------------|-------|------|------|--------|------|
| SigLIP <sup>1</sup> | B/16  | B    | 16 k | 16     | 3    |
| SigLIP              | B/16  | B    | 32 k | 32     | 2    |
| SigLIP              | B/16  | B    | 32 k | 32     | 5    |

## CLIP (2021)

Batch size: 32

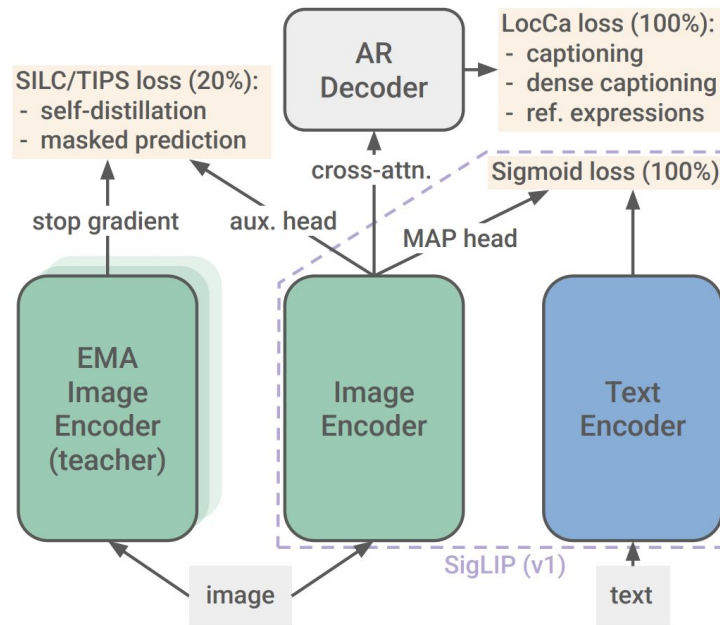
the largest Vision Transformer took 12 days on 256 V100 GPUs. For the ViT-L/14 we also pre-train at a higher 336

<sup>1</sup> [https://arxiv.org/pdf/2305.01989v1.pdf](#)

# SigLIP2: The 2025 era

Builds on **SigLIP** but uses multiple **complementary losses**:

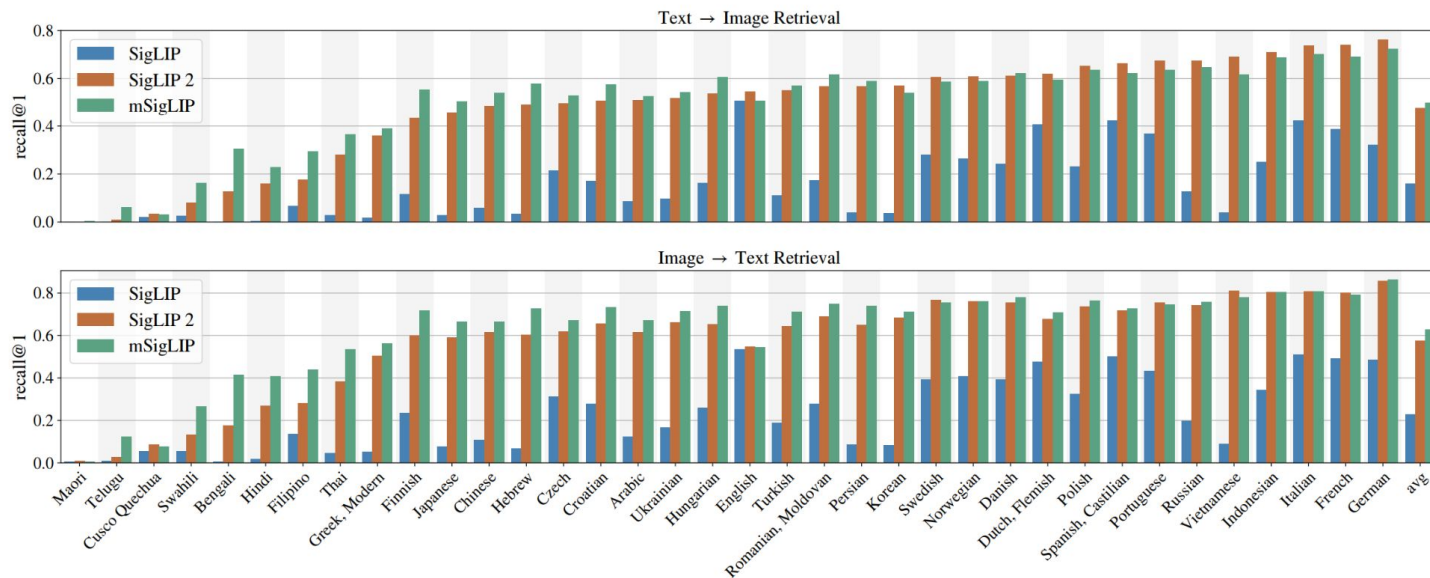
- Sigmoid contrastive loss (global alignment)
- Self-supervised image loss (SILC / TIPS)
- Autoregressive decoder loss (LocCa)



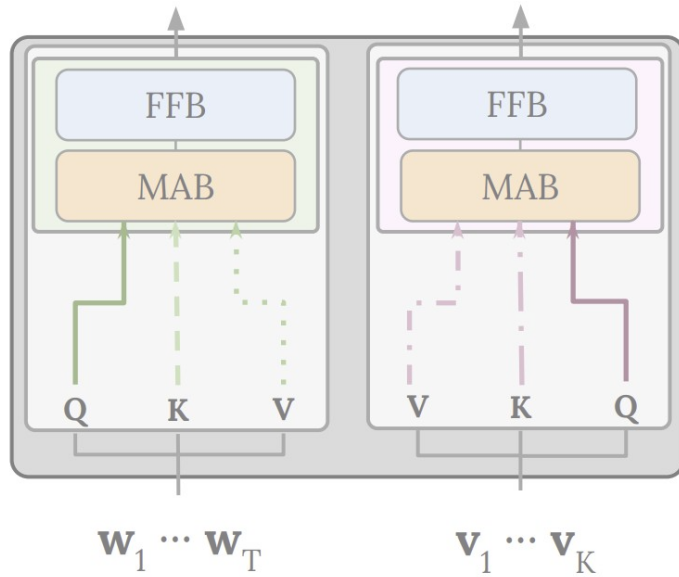


# SigLIP2: The 2025 era

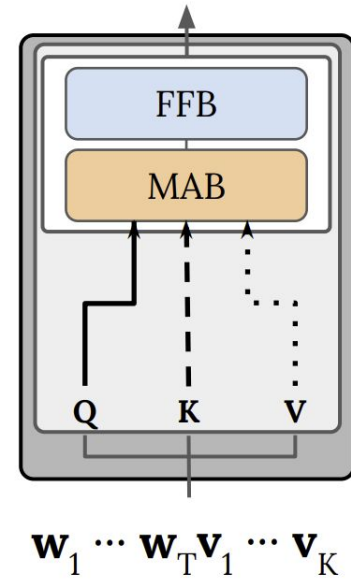
SigLIP2 is uses a single **shared text encoder** trained on **many languages** at once.



## Single Modality (no fusion)



## Single-stream Transformer (early fusion)

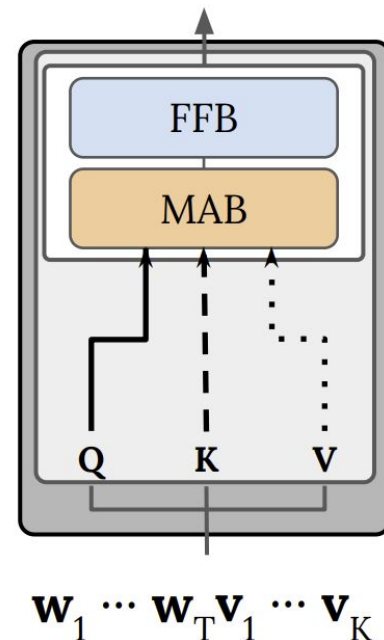


# Early Fusion: One Single Encoder

Both modalities are combined at **input level** by **concatenating** both text and image features.

Allows the model to learn **cross-modal interactions** from the beginning by using **one single encoder**

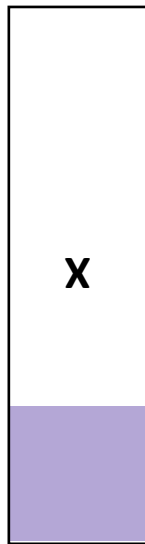
**Attention** can be computed over both modalities!



MAB = Multi-Head Attention Block  
FFB = Feed-Forward Block

# Scaled Dot-Product Attention

What is in the image?

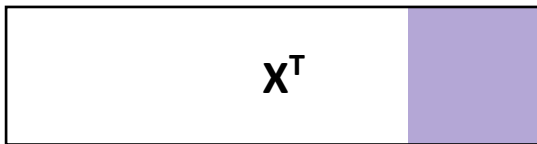


$$\cdot \mathbf{W}_Q =$$

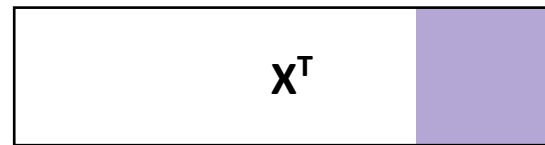


$$\mathbf{K} = \mathbf{X} * \mathbf{W}_K$$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$$



$$\cdot \mathbf{W}_K =$$



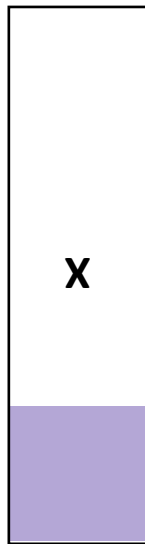
$$\cdot \mathbf{W}_V =$$



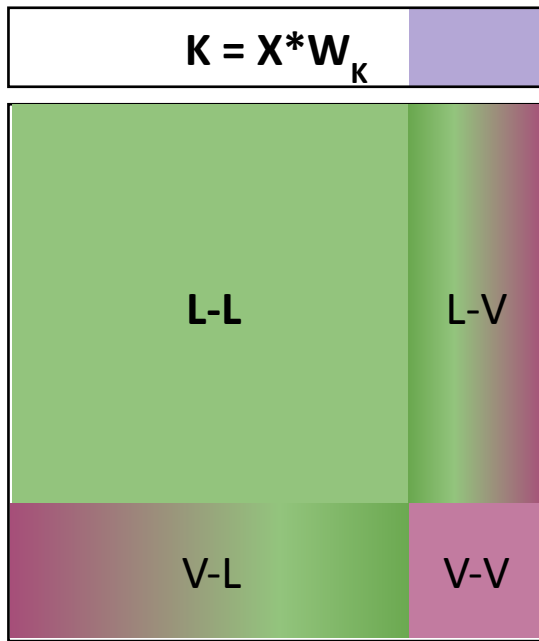
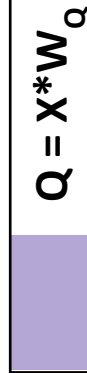
.

# Scaled Dot-Product Attention

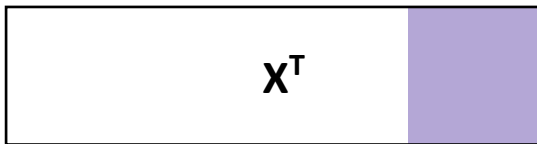
What is in the image?



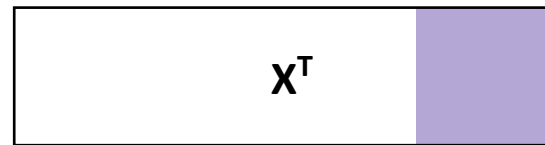
$$\cdot \quad \mathbf{W}_Q =$$



$$\mathbf{K} = \mathbf{X} * \mathbf{W}_K$$



$$\cdot \quad \mathbf{W}_K =$$



$$\cdot \quad \mathbf{W}_V =$$



.

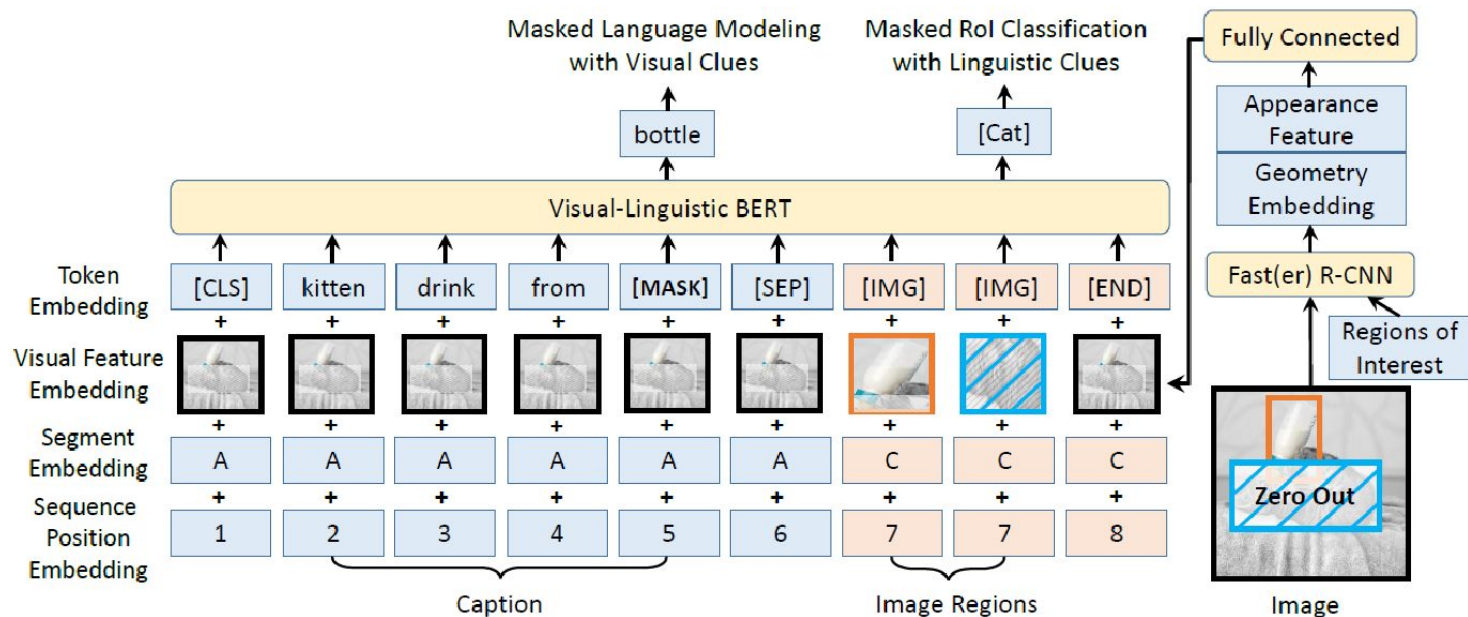
# Early Fusion: VL BERT

An extension of the original BERT model designed to **jointly** process visual and language data

## Pre-training tasks:

- **Masked Language Modeling** - It uses the **multimodal context** to predict the masked word
- **Masked Object Classification** - Similar to MLM, a region of the image is masked (i.e., A bounding box of a dog)

# Early Fusion: VL BERT



# Early Fusion: Is it better?

## Single Modality (no fusion)



Efficient Training



Preserves independent  
modality representations



Limited cross-modal  
interaction

## Single-stream Transformer (early fusion)



Enables early cross-modal  
interactions (joint attention)



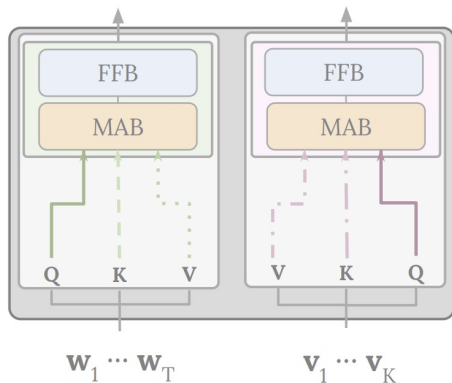
Ideal for tasks requiring deep  
vision-language reasoning



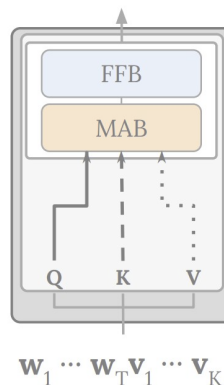
Computationally expensive



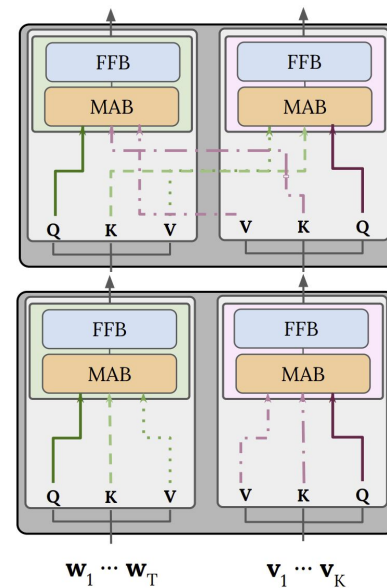
## Single Modality (no fusion)



## Single-stream Transformer (early fusion)



## Dual-stream Transformer (late fusion)

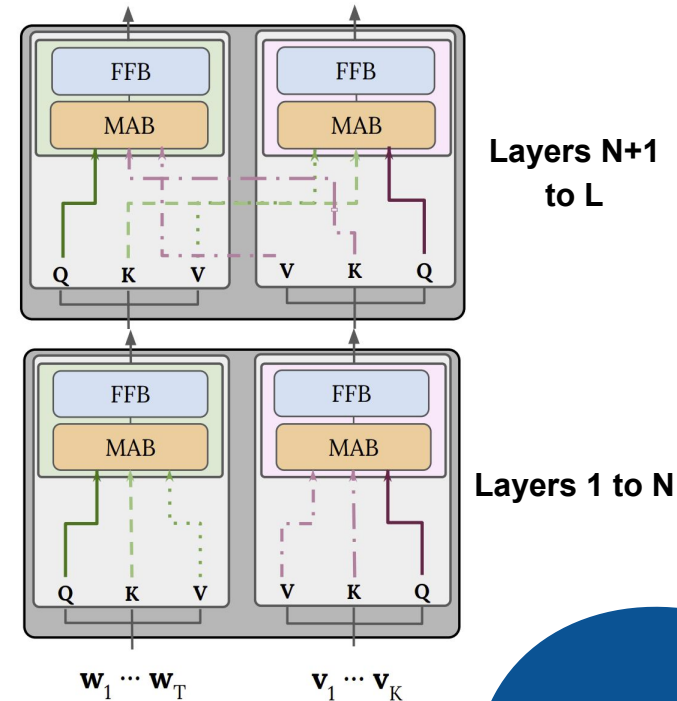


# Late Fusion: Two types of Layers

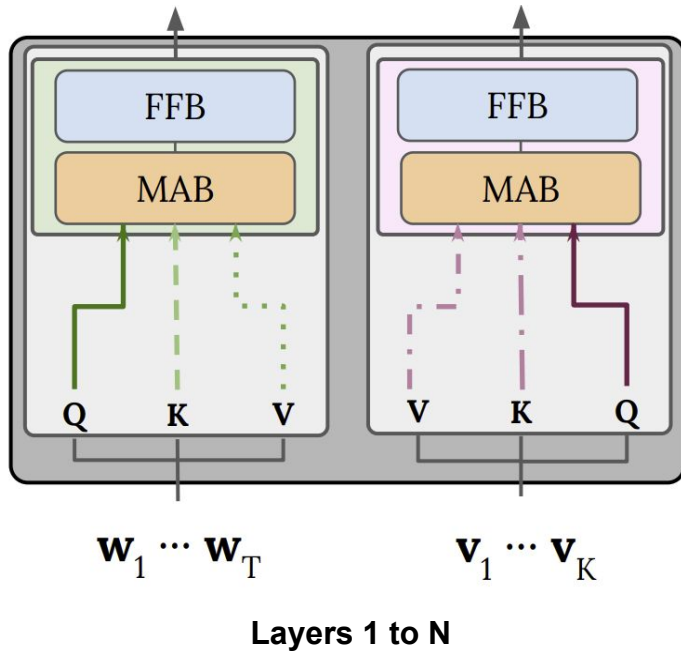
The **dual stream** approaches involves two different types of transformer layers:

**Inter-model Layers:** Embeddings from different modalities are exchanged across streams

**Intra-model Layers:** Separate streams for language and vision

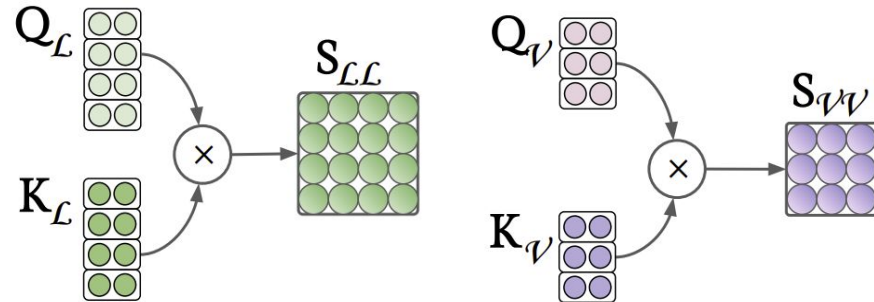


# Late Fusion: Intra-modal



Does this look familiar?

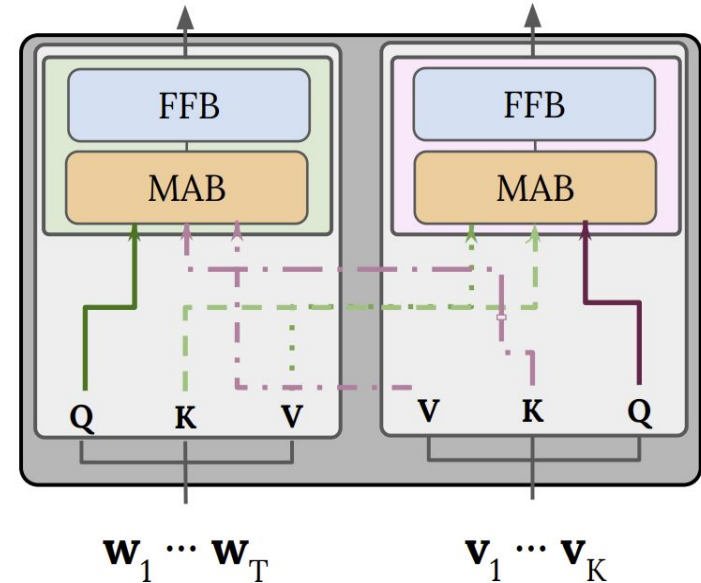
Single Modality  
(no fusion)



# Late Fusion: Inter-modal

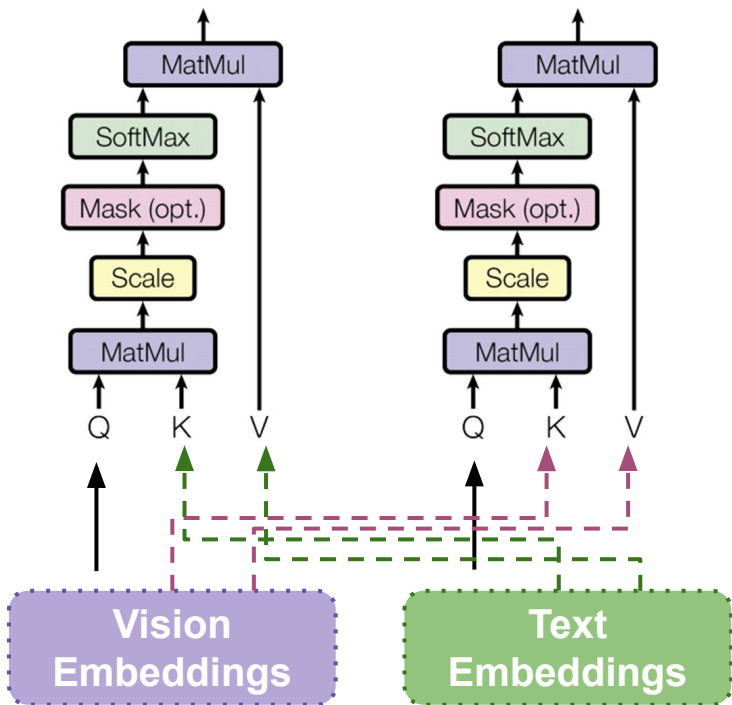
These layers **explicitly** model cross-modal interaction via a **cross-modal attention module**

Co-attention mechanism



MAB = Multi-Head Attention Block  
FFB = Feed-Forward Block

# Co-attention Mechanism: Inter-modal



The **keys** and **values** from each modality are used as input to the other modality's multi-headed attention block.

$$\text{Attention}(Q_v, K_w, V_w) = \text{softmax}\left(\frac{Q_v \cdot K_w}{\sqrt{d_k}}\right) \cdot V_w$$

$$\text{Attention}(Q_w, K_v, V_v) = \text{softmax}\left(\frac{Q_w \cdot K_v}{\sqrt{d_k}}\right) \cdot V_v$$

# Co-attention Mechanism: Inter-modal

The attention block produces **attention-pooled features** for each modality conditioned on the other:

1. Language attention is conditioned in the visual stream

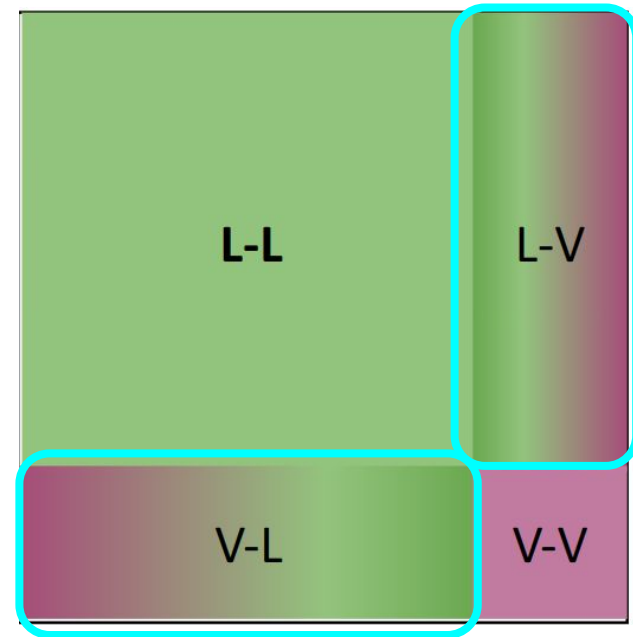
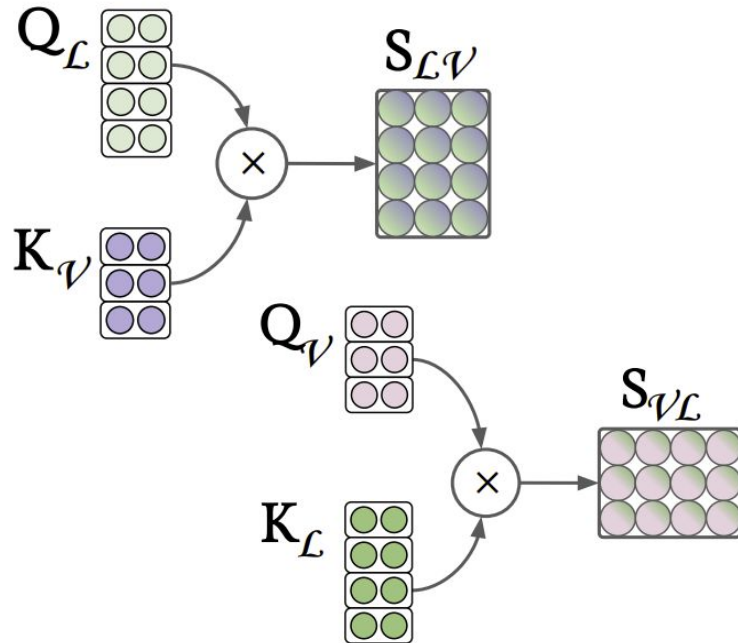
**Keys** and **Values** come from **Image embeddings**

2. Image attention is conditioned in the linguistic stream

**Keys** and **Values** come from **Text embeddings**

This mechanism ensures that each modality learns context-aware representations by leveraging information from the other modality!

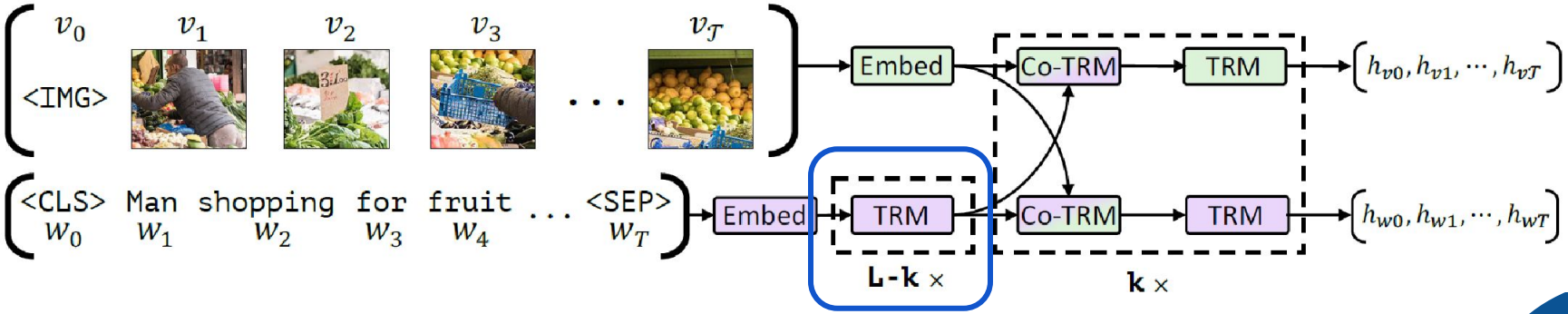
# Multimodal Co-Attention



# Late Fusion: ViLBERT

This model leverages **two separate streams** with the modalities interacting through **co-attention layers**.

There's also a difference in **processing** depth between both modalities (vision involves fairly more high-level features)



TRM = Transformer Blocks

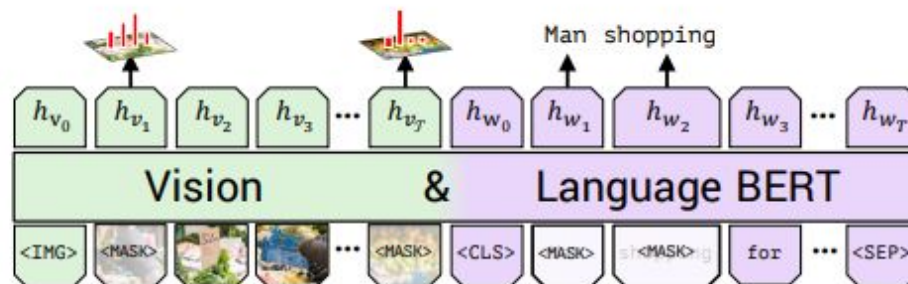


# ViLBERT: Pre-training

**Masked Multimodal Learning:** Similar to the standard MLM but incorporates both vision and language.

**Masking text tokens** in a sentence and predicting them using visual context and surrounding text.

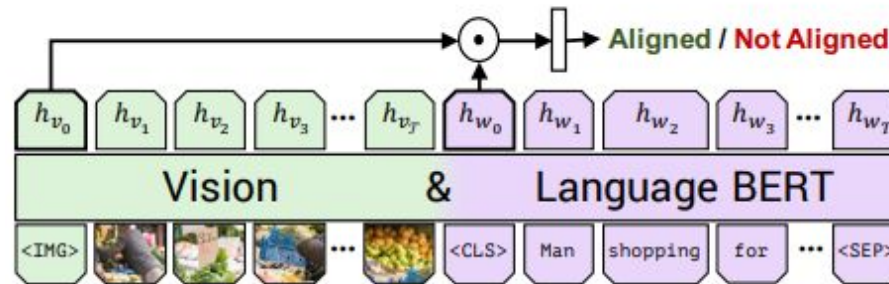
**Masking image regions** and predicting their categories using text context and visual information



(a) Masked multi-modal learning

# ViLBERT: Pre-training

**Multimodal-Alignment Prediction:** The model predicts whether or not the caption describes the image content (i.e., whether the text and the image are **aligned** or not)



(b) Multi-modal alignment prediction

### Single Modality (no fusion)



Efficient Training



Preserves independent  
modality  
representations



Limited cross-modal  
interaction

### Single-stream Transformer (early fusion)



Enables early  
cross-modal interactions  
(joint attention)



Ideal for tasks requiring  
deep vision-language  
reasoning



Computationally  
expensive

### Dual-stream Transformer (late fusion)



Explicit cross-modal  
alignment



Different processing  
needs



Higher complexity

# Exercise Time

